

$$= (1-t + (-1) \cdot t) \cdot p + qt \quad (\text{Algebraic property viii})$$

$$= (1-t) \cdot p + qt \quad (\text{Algebraic property vi}) \quad \square$$

1. Let 1 b) The line segment from p to q is the set of points of the form $(1-t)p + tq$ for $0 \leq t \leq 1$ (as show in the figure below). Show that a linear transformation T maps this line segment onto a line segment or onto a single point.

$$(t=1) \quad \bullet \quad (1-t)p + tq$$

2. (iv)

Solution: Let x be a point on the line segment from p and q , $p(t=0)$

$$\therefore x = (1-t)p + tq \quad \text{for some } t \in [0, 1].$$

$$T(x) = T((1-t)p + tq) = T((1-t)p) + T(tq) \quad (\text{Condition 1})$$

$$= (1-t)T(p) + tT(q) \quad (\text{Condition 2})$$

Let $T(p)$ and $T(q)$ be linearly dependent. $\therefore$ Atleast one must be a scalar multiple of the other.

WLOG, let $T(q)$ be a scalar multiple of $T(p)$. $\therefore \exists x \in \mathbb{R}$ s.t. $T(q) = xT(p)$

$$T(x) = (1-t)T(p) + tT(q) \Rightarrow T(x) = T(p)(1-t + tx) + 0.$$

$\therefore T(x)$ is a line through the origin in the direction of $T(p)$.

If $T(p)$ and $T(q)$ are linearly independent, then $T(p)$ and $T(q)$ are not scalar multiples of one another.

$\therefore T(x)$ is a line segment from $T(p)$ to $T(q)$. $\square$

28) Let u and v be vectors in $\mathbb{R}^n$. It can be shown that the set P of all points in the parallelogram determined by u and v has the form $au + bv$, for $0 \leq a, b \leq 1$. Let $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ be a linear transformation. Explain why the image of a point in P under the transformation T lies in the parallelogram determined by $T(u)$ and $T(v)$.

Solution: Let $x \in P$. $\therefore x = au + bv$, for some a, b in $[0, 1]$.

$$T(x) = T(au + bv) = T(au) + T(bv) \quad (\text{Condition 1}) = aT(u) + bT(v) \quad (\text{Condition 2})$$

$\therefore a, b \in [0, 1]$. By given condition, $T(x)$ is a point on the $\parallel\text{gm}$ determined by $T(u)$ and $T(v)$. $\square$